CCIT Report No. 353

Multi-Cell Uplink Spectral Efficiency of Coded DS-CDMA with Random Signatures

Benjamin M. Zaidel, Shlomo Shamai (Shitz) and Sergio Verdú

ABSTRACT

A simple multi-cell uplink communication model is suggested and analyzed for optimally coded randomly spread DS-CDMA. The model adheres to Wyner"s (1994) infinite linear cell-array model, according to which only adjacent-cell interference is present, and characterized by a *single* parameter $0 \le \alpha \le 1$. The discussion is confined to asymptotic analysis where both the number of users and the processing gain go to infinity, while their ratio goes to some finite constant. Single cell-site processing is assumed and four multiuser detection strategies are considered: the matched-filter detector, "optimum" detection with adjacent-cell interference treated as Gaussian noise, the linear MMSE detector, and a detector that performs MMSE based successive interference cancellation for intra-cell users with linear MMSE processing of adjacent-cell interference. Spectral efficiency is evaluated under three power allocation policies: equal received powers (for all users), equal rates, and a maximal spectral efficiency policy. Comparative results demonstrate how performance is affected by the introduction of inter-cell interference, and what is the penalty associated with the randomly spread coded DS-CDMA strategy. Finally, the effect of inter-cell time-sharing protocols as suggested by Shamai and Wyner (1997) is also examined, and a significant system performance enhancement is observed.